

2.1 核心板资源描述

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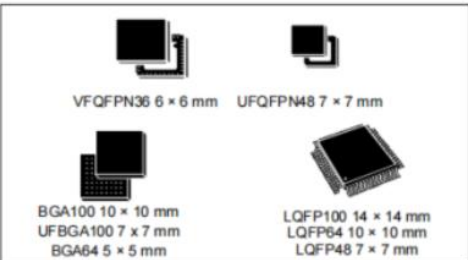
STM32F103x8
STM32F103xB

Medium-density performance line Arm[®]-based 32-bit MCU with
64 or 128 KB Flash, USB, CAN, 7 timers, 2 ADCs, 9 com. interfaces

Datasheet - production data

Features

- Arm[®] 32-bit Cortex[®]-M3 CPU core
 - 72 MHz maximum frequency, 1.25 DMIPS / MHz (Dhrystone 2.1) performance at 0 wait state memory access
 - Single-cycle multiplication and hardware division
- Memories
 - 64 or 128 Kbytes of Flash memory
 - 20 Kbytes of SRAM
- Clock, reset and supply management
 - 2.0 to 3.6 V application supply and I/Os
 - POR, PDR, and programmable voltage detector (PVD)
 - 4 to 16 MHz crystal oscillator
 - Internal 8 MHz factory-trimmed RC
 - Internal 40 kHz RC
 - PLL for CPU clock
 - 32 kHz oscillator for RTC with calibration
- Low-power
 - Sleep, Stop and Standby modes
 - V_{BAT} supply for RTC and backup registers
- 2x 12-bit, 1 μ s A/D converters (up to 16 channels)



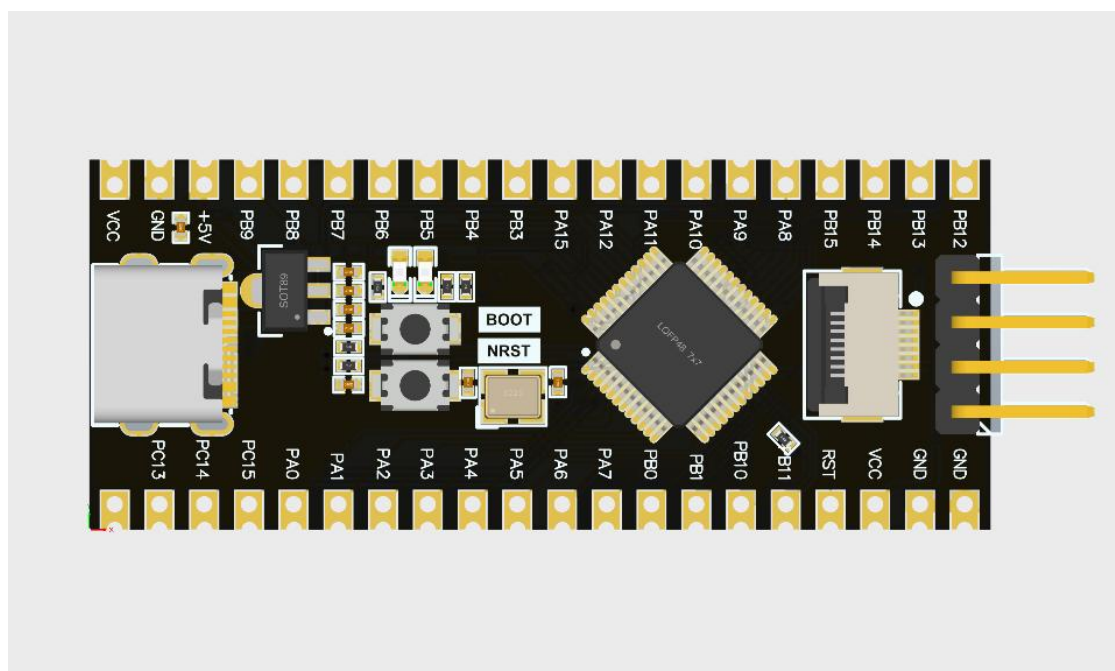
- Debug mode
 - Serial wire debug (SWD) and JTAG interfaces
- Seven timers
 - Three 16-bit timers, each with up to 4 IC/OC/PWM or pulse counter and quadrature (incremental) encoder input
 - 16-bit, motor control PWM timer with dead-time generation and emergency stop
 - Two watchdog timers (independent and window)
 - SysTick timer 24-bit downcounter
- Up to nine communication interfaces
 - Up to two I²C interfaces (SMBus/PMBus[®])
 - Up to three USARTs (ISO 7816 interface, LIN, IrDA capability, modem control)
 - Up to two SPIs (18 Mbit/s)

ARM 32 位 Cortex-M3 微控制器, 72MHz 主频, 64kB 闪存, 20kB SRAM, PLL, 嵌入式内部 RC 8MHz 和低速时钟 32kHz, 实时时钟, 嵌套的中断控制器, 省电模式, JTAG 和 SWD, 与输入捕捉、输出比较和 PWM 同步的 16 位定时器, 16 位 6-ch 的高级计时器, 2 个 16 位的看门狗定时器, 滴答定时器, 2 个 SPI, 2 个 I2C, 3 个 USART, USB 2.0 的全速度接口, CAN 接口 (2.0B 主动), 2 个 12 位 10-ch A/D 转换器, 高速输入/输出端口。

STM32F103C8T6 的命名含义, 如下:

STM32型号的说明:		
以STM32F103C8T6这个型号的芯片为例,该型号的组成为7个部分,其命名规则如下:		
1	STM32	STM32代表ARM Cortex-M内核的32位微控制器。
2	F	F代表芯片子系列
3	103	103代表增强型系列
4	C	C这一项代表引脚,其中T代表36脚, C代表48, R代表64, V代表100脚, Z代表144, I代表176脚
5	8	8这一项代表内Flash容量,其中6代表32K字节Flash, 8代表64K字节Flash, B代表128K字节Flash, C代表256K字节Flash, D代表384K字节Flash, E代表512K字节Flash, G代表1M字节Flash
6	T	T这一项代表封装,其中H代表BGA装,T代表LQFP封装,U代表VFQFPN封装
7	6	6这一项代表工作温度范围,其中6代表-40—85℃, 7代表-40—105℃

我们开发板使用的是 STM32F103C8T6 芯片，芯片外观及引脚图如下：



芯片封装图如下:

Figure 8. STM32F103xx performance line LQFP48 pinout

